Ex:No:08

Reverse shell using TCP Socket

Aim:

To develop a TCP Socket based client Server program for communication between two systems.

Procedure:-

* Import the Socket module.
* Create a TCP Socket 127.0.0.1 and Port 5000.
* Start the server and wait for a client Connection.
* Connect the client to server.
* Send and receive to server message using send() and recv().
* Terminate the connection using exit.

Result:

Thus, a TCP Socket based client-Server Communication program was successfully executed on localhost.

```
import socket

SERVER_IP = "127.0.0.1"
PORT = 5000

server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind((SERVER_IP, PORT))
server.listen(1)

print("Waiting for client...")
conn, addr = server.accept()
print("Connected:", addr)

while True:
    message = conn.recv(1024).decode()

    if not message or message.lower() == "exit":
        break

    print("Client:", message)
    response = input("Enter response: ")

    conn.send(response.encode())

conn.close()
server.close()
```

client.py - C:\Users\Admin\Desktop\exp8\client.py (3.12.10)

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```
import socket

SERVER_IP = "127.0.0.1"
PORT = 5000

client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect((SERVER_IP, PORT))

print("Connected to server")

while True:
    message = input("Enter message: ")

    if message.lower() == "exit":
        break

    client.send(message.encode())
    response = client.recv(1024).decode()
    print("Server:", response)

client.close()
```

C:\Windows\system32\cmd.exe - python client.py

```
Microsoft Windows [Version 10.0.19045.6466]
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C:\Users\Admin>cd Desktop\exp8

C:\Users\Admin\Desktop\exp8>python client.py
Connected to server
Enter message: hello
Server: hello client
Enter message:
```

C:\Windows\system32\cmd.exe - python server.py

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>cd Desktop\exp8

C:\Users\Admin\Desktop\exp8>python server.py
Waiting for client...
Connected: ('127.0.0.1', 63293)
Client: hello
Enter response: hello client
```